WEB TECHNOLOGY ASSIGNMENT - 2

Name: Gokula Krishnan R

Roll No: 24CS068

Class : III-CSE-B

Course Code : U21CS501

Course Title : Web Technology

Date: 17/10/2026

QUESTION 1

Question: Create HTML forms for student registration (name, roll number, email, course, file upload). Apply CSS styles for layout, colors and responsive design. Implement client-side validation (JavaScript) for required fields, email format and password strength.

Code

File: index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Registration</title>

    <!-- External CSS file -->
    <link rel="stylesheet" href="style.css">
</head>

<body>

    <h2>Student Registration Form</h2>

    <form id="registrationForm">

        <label>Student Name:</label>
        <input type="text" id="name">
        <p id="nameError"></p>

        <label>Roll Number:</label>
        <input type="text" id="rollno">
        <p id="rollnoError"></p>

        <label>Email:</label>
        <input type="email" id="email">
        <p id="emailError"></p>

        <label>Course:</label>
        <select id="course">
            <option value="">Select Course</option>
            <option value="CSE">CSE</option>
            <option value="ECE">ECE</option>
            <option value="EEE">EEE</option>
            <option value="IT">IT</option>
        </select>
        <p id="courseError"></p>

        <label>Password:</label>
        <input type="password" id="password">
        <p id="passwordError"></p>

        <label>Upload File:</label>
        <input type="file" id="file">

        <button type="submit">Register</button>

    </form>

    <!-- External JavaScript file -->
    <script src="script.js"></script>

</body>
</html>
```

File: style.css

```
body {
```

```
    font-family: "Trebuchet MS", Helvetica, sans-serif;
    background-color: #e8f3f1;
    margin: 0;
    padding: 20px;
}

h2 {
    text-align: center;
    color: #134e4a;
}

form {
    max-width: 500px;
    margin: auto;
    background-color: white;
    padding: 25px;
    border-radius: 16px;
    box-shadow: 0 8px 24px #0f3d3a26;
}

label {
    display: block;
    margin-top: 15px;
    margin-bottom: 5px;
    font-weight: bold;
}

input,
select {
    width: 100%;
    padding: 10px;
    box-sizing: border-box;
    border: 1px solid #8cc5bd;
    border-radius: 10px;
}

button {
    width: 100%;
    padding: 10px;
    margin-top: 20px;
    background-color: #0f766e;
    color: white;
    border: none;
    border-radius: 10px;
    cursor: pointer;
}

button:hover {
    background-color: #115e59;
}

p {
    color: #c2410c;
    font-size: 14px;
    margin: 5px 0;
}

/* Responsive design */
@media (max-width: 600px) {
    form {
        padding: 15px;
    }

    body {
        padding: 10px;
    }
}
```

File: script.js

```
document.getElementById("registrationForm").addEventListener("submit", function(event) {

    event.preventDefault();

    let name = document.getElementById("name").value.trim();
    let rollno = document.getElementById("rollno").value.trim();
    let email = document.getElementById("email").value.trim();
    let course = document.getElementById("course").value;
    let password = document.getElementById("password").value;

    let valid = true;

    // Clear previous error messages
    document.getElementById("nameError").innerHTML = "";
    document.getElementById("rollnoError").innerHTML = "";
    document.getElementById("emailError").innerHTML = "";
    document.getElementById("courseError").innerHTML = "";
    document.getElementById("passwordError").innerHTML = "";

    // Required field validation
    if (name === "") {
        document.getElementById("nameError").innerHTML = "Name is required";
        valid = false;
    }

    if (rollno === "") {
        document.getElementById("rollnoError").innerHTML = "Roll number is required";
        valid = false;
    }

    if (email === "") {
        document.getElementById("emailError").innerHTML = "Email is required";
        valid = false;
    }

    if (course === "") {
        document.getElementById("courseError").innerHTML = "Please select a course";
        valid = false;
    }

    if (password === "") {
        document.getElementById("passwordError").innerHTML = "Password is required";
        valid = false;
    }

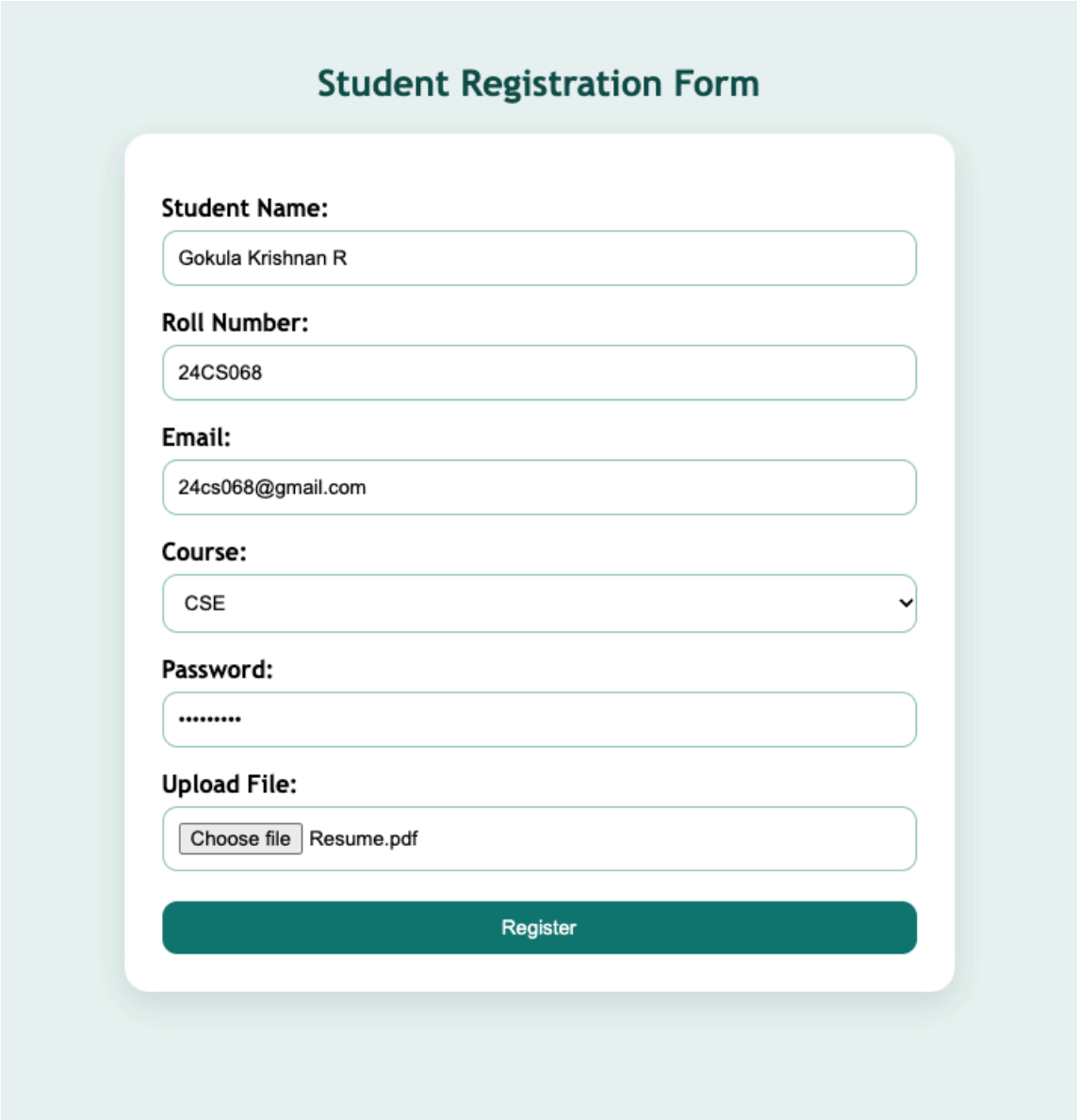
    // Email format validation
    let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

    if (email !== "" && !emailPattern.test(email)) {
        document.getElementById("emailError").innerHTML = "Enter a valid email";
        valid = false;
    }

    // Password strength validation
    if (password !== "" && password.length < 8) {
        document.getElementById("passwordError").innerHTML =
            "Password must contain at least 8 characters";
        valid = false;
    }

    // Successful submission
    if (valid) {
        alert("Registration successful!");
    }
});
```

Output


Student Registration Form
Student Name:
Gokula Krishnan R
Roll Number:
24CS068
Email:
24cs068@gmail.com
Course:
CSE
Password:
Upload File:
Choose file
Resume.pdf
Register

QUESTION 2

Question: Create PHP pages for CRUD operations (Add, Edit, Delete, View) for student details using MySQL; handle profile image/document uploads; store file metadata; display uploaded files; validate file size/type and prevent overwriting.

Code

File: add.php

```
<?php
include "db.php";

if (isset($_POST["submit"])) {

    $name = $_POST["name"];
    $roll_no = $_POST["roll_no"];
    $course = $_POST["course"];

    $sql = "INSERT INTO students (name, roll_no, course)
            VALUES ('$name', '$roll_no', '$course')";

    if ($conn->query($sql) === TRUE) {
        echo "Student added successfully!";
    } else {
        echo "Error: " . $conn->error;
    }
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Add Student</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

<h2>Add Student</h2>

<form method="POST">

    <label>Student Name:</label>
    <input type="text" name="name" required>

    <label>Roll Number:</label>
    <input type="text" name="roll_no" required>

    <label>Course:</label>
    <input type="text" name="course" required>

    <button type="submit" name="submit">Add Student</button>

</form>

<br>

<a href="index.php">Back to Student List</a>

</body>
</html>
```

File: db.php

```
<?php
```

```php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "student_db";

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
    die("Database connection failed: " . $conn->connect_error);
}

?>
```

File: delete.php

```php
<?php
include "db.php";

$id = $_GET["id"];

$sql = "DELETE FROM students WHERE id = $id";

if ($conn->query($sql) === TRUE) {
    echo "Student deleted successfully!";
} else {
    echo "Error: " . $conn->error;
}

echo "<br><br>";
echo "<a href='index.php'>Back to Student List</a>";

?>
```

File: edit.php

```php
<?php
include "db.php";

$id = $_GET["id"];

// Get student details
$sql = "SELECT * FROM students WHERE id = $id";
$result = $conn->query($sql);
$student = $result->fetch_assoc();

// Update student details
if (isset($_POST["update"])) {

    $name = $_POST["name"];
    $roll_no = $_POST["roll_no"];
    $course = $_POST["course"];

    $sql = "UPDATE students
            SET name='$name', roll_no='$roll_no', course='$course'
            WHERE id=$id";

    if ($conn->query($sql) === TRUE) {
        echo "Student updated successfully!";
    } else {
        echo "Error: " . $conn->error;
    }
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Edit Student</title>
    <link rel="stylesheet" href="style.css">
</head>
```

```
<body>

<h2>Edit Student</h2>

<form method="POST">

    <label>Student Name:</label>
    <input type="text" name="name"
           value="<?php echo $student['name']; ?>" required>

    <label>Roll Number:</label>
    <input type="text" name="roll_no"
           value="<?php echo $student['roll_no']; ?>" required>

    <label>Course:</label>
    <input type="text" name="course"
           value="<?php echo $student['course']; ?>" required>

    <button type="submit" name="update">
        Update Student
    </button>

</form>

<br>

<a href="index.php">Back to Student List</a>

</body>
</html>
```

File: index.php

```
<?php
include "db.php";

$sql = "SELECT * FROM students";
$result = $conn->query($sql);
?>

<!DOCTYPE html>
<html>
<head>
    <title>Student Admin Panel</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

<h2>Student Admin Panel</h2>

<a href="add.php">Add Student</a>

<br><br>

<table border="1">

    <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Roll Number</th>
        <th>Course</th>
        <th>Action</th>
    </tr>

    <?php while ($row = $result->fetch_assoc()) { ?>

    <tr>
```

```
        <td><?php echo $row["id"]; ?></td>

        <td><?php echo $row["name"]; ?></td>

        <td><?php echo $row["roll_no"]; ?></td>

        <td><?php echo $row["course"]; ?></td>

        <td>
            <a href="view.php?id=<?php echo $row["id"]; ?>">View</a>

            <a href="edit.php?id=<?php echo $row["id"]; ?>">Edit</a>

            <a href="delete.php?id=<?php echo $row["id"]; ?>">Delete</a>
        </td>

    </tr>

    <?php } ?>

</table>

</body>
</html>
```

File: upload.php

```
<?php
include "db.php";

$student_id = $_GET["id"];

if (isset($_POST["upload"])) {

    $file_name = $_FILES["file"]["name"];
    $file_type = $_FILES["file"]["type"];
    $file_size = $_FILES["file"]["size"];
    $file_tmp = $_FILES["file"]["tmp_name"];

    // Allowed file types
    $allowed_types = [
        "image/jpeg",
        "image/png",
        "application/pdf"
    ];

    // Maximum file size: 2 MB
    $max_size = 2 * 1024 * 1024;

    if (!in_array($file_type, $allowed_types)) {

        echo "Only JPG, PNG and PDF files are allowed.";

    } elseif ($file_size > $max_size) {

        echo "File size must be less than 2 MB.";

    } else {

        // Create a unique file name
        $new_file_name = time() . "_" . $file_name;

        $file_path = "uploads/" . $new_file_name;

        // Move file to uploads folder
        if (move_uploaded_file($file_tmp, $file_path)) {

            $sql = "INSERT INTO files
                    (student_id, file_name, file_path, file_type)
                    VALUES
```

```
                    ('$student_id', '$file_name', '$file_path', '$file_type')";

            if ($conn->query($sql) === TRUE) {
                echo "File uploaded successfully!";
            } else {
                echo "Database error: " . $conn->error;
            }

        } else {

            echo "File upload failed.";

        }
    }
}
?>

<!DOCTYPE html>
<html>

<head>
    <title>Upload File</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

<h2>Upload Student File</h2>

<form method="POST" enctype="multipart/form-data">

    <label>Select File:</label>

    <input type="file" name="file" required>

    <button type="submit" name="upload">
        Upload File
    </button>

</form>

<br>

<a href="view.php?id=<?php echo $student_id; ?>">
    Back to Student Details
</a>

</body>

</html>
```

File: view.php

```
<?php
include "db.php";

$id = $_GET["id"];

// Get student details
$sql = "SELECT * FROM students WHERE id = $id";
$result = $conn->query($sql);
$student = $result->fetch_assoc();

// Get uploaded files
$file_sql = "SELECT * FROM files WHERE student_id = $id";
$file_result = $conn->query($file_sql);
?>

<!DOCTYPE html>
<html>
```

```
<head>
    <title>View Student</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

<h2>Student Details</h2>

<p>
    <b>Name:</b>
    <?php echo $student["name"]; ?>
</p>

<p>
    <b>Roll Number:</b>
    <?php echo $student["roll_no"]; ?>
</p>

<p>
    <b>Course:</b>
    <?php echo $student["course"]; ?>
</p>

<hr>

<h3>Uploaded Files</h3>

<a href="upload.php?id=<?php echo $id; ?>">
    Upload New File
</a>

<br><br>

<?php

if ($file_result->num_rows > 0) {

    while ($file = $file_result->fetch_assoc()) {
?>

    <div>

        <p>
            <b>File Name:</b>
            <?php echo $file["file_name"]; ?>
        </p>

        <p>
            <b>File Type:</b>
            <?php echo $file["file_type"]; ?>
        </p>

        <p>
            <b>Upload Date:</b>
            <?php echo $file["upload_date"]; ?>
        </p>

        <?php if (strpos($file["file_type"], "image") !== false) { ?>

            <img src="<?php echo $file["file_path"]; ?>"
                 width="200">

        <?php } else { ?>

            <a href="<?php echo $file["file_path"]; ?>"
               target="_blank">
                View / Download File
            </a>
```

```
        <?php } ?>

        <hr>

    </div>

<?php
    }

} else {

    echo "No files uploaded yet.";

}
?>

<br>

<a href="index.php">Back to Student List</a>

</body>

</html>
```

File: style.css

```
/* ------------------------------
   General Page Styling
------------------------------ */

body {
    font-family: "Trebuchet MS", Helvetica, sans-serif;
    background: linear-gradient(135deg, #e6f4f1, #f7fbfa);
    margin: 0;
    padding: 30px;
    color: #1f3a3a;
}

/* ------------------------------
   Headings
------------------------------ */

h2 {
    text-align: center;
    color: #134e4a;
    margin-bottom: 25px;
    font-size: 30px;
}

h3 {
    color: #115e59;
    margin-top: 25px;
}

/* ------------------------------
   Forms
------------------------------ */

form {
    width: 420px;
    max-width: 90%;
    margin: 30px auto;
    background-color: white;
    padding: 30px;
    border-radius: 15px;
    box-shadow: 0 8px 25px rgba(0, 0, 0, 0.12);
}

label {
```

```
    display: block;
    margin-top: 15px;
    margin-bottom: 7px;
    font-weight: bold;
    color: #115e59;
}

input {
    width: 100%;
    padding: 12px;
    box-sizing: border-box;
    border: 1px solid #b3d6d0;
    border-radius: 8px;
    font-size: 15px;
    outline: none;
    transition: 0.3s;
}

input:focus {
    border-color: #14b8a6;
    box-shadow: 0 0 5px rgba(20, 184, 166, 0.3);
}

input[type="file"] {
    background-color: #f0faf8;
}

/* ------------------------------
   Buttons
------------------------------ */

button {
    width: 100%;
    padding: 12px;
    margin-top: 22px;
    background: linear-gradient(135deg, #14b8a6, #0f766e);
    color: white;
    border: none;
    border-radius: 8px;
    cursor: pointer;
    font-size: 16px;
    font-weight: bold;
    transition: 0.3s;
}

button:hover {
    transform: translateY(-2px);
    box-shadow: 0 5px 15px rgba(15, 118, 110, 0.3);
}

/* ------------------------------
   Links
------------------------------ */

a {
    color: #0f766e;
    text-decoration: none;
    font-weight: bold;
    margin-right: 12px;
}

a:hover {
    color: #134e4a;
    text-decoration: underline;
}

/* Add Student / Upload Links */

body > a {
    display: inline-block;
```

```
    background-color: #0f766e;
    color: white;
    padding: 10px 18px;
    border-radius: 8px;
    margin-bottom: 10px;
    text-decoration: none;
}

body > a:hover {
    background-color: #115e59;
    color: white;
    text-decoration: none;
}

/* ------------------------------
   Student Table
------------------------------ */

table {
    width: 95%;
    margin: 25px auto;
    border-collapse: separate;
    border-spacing: 0;
    background-color: white;
    border-radius: 16px;
    overflow: hidden;
    box-shadow: 0 8px 25px rgba(0, 0, 0, 0.10);
}

th {
    background: linear-gradient(135deg, #0f766e, #14b8a6);
    color: white;
    padding: 15px;
    font-size: 15px;
}

td {
    padding: 13px;
    text-align: center;
    border-bottom: 1px solid #d5e8e4;
}

tr:hover {
    background-color: #effaf8;
}

tr:last-child td {
    border-bottom: none;
}

/* ------------------------------
   Student Details
------------------------------ */

body > p {
    background-color: white;
    width: 600px;
    max-width: 90%;
    margin: 10px auto;
    padding: 12px 18px;
    border-radius: 8px;
    box-shadow: 0 3px 10px rgba(0, 0, 0, 0.06);
}

/* ------------------------------
   Uploaded Files
------------------------------ */

div {
    background-color: white;
```

```
    padding: 20px;
    margin: 20px auto;
    max-width: 600px;
    border-radius: 12px;
    box-shadow: 0 5px 18px rgba(0, 0, 0, 0.08);
}

div p {
    margin: 8px 0;
}

img {
    display: block;
    max-width: 250px;
    max-height: 250px;
    margin: 15px auto;
    border-radius: 10px;
    border: 2px solid #d5e8e4;
}

/* ------------------------------
   Horizontal Line
------------------------------ */

hr {
    border: none;
    border-top: 1px solid #d5e8e4;
    margin: 25px 0;
}

/* ------------------------------
   Responsive Design
------------------------------ */

@media (max-width: 600px) {

    body {
        padding: 15px;
    }

    h2 {
        font-size: 24px;
    }

    form {
        width: 90%;
        padding: 20px;
    }

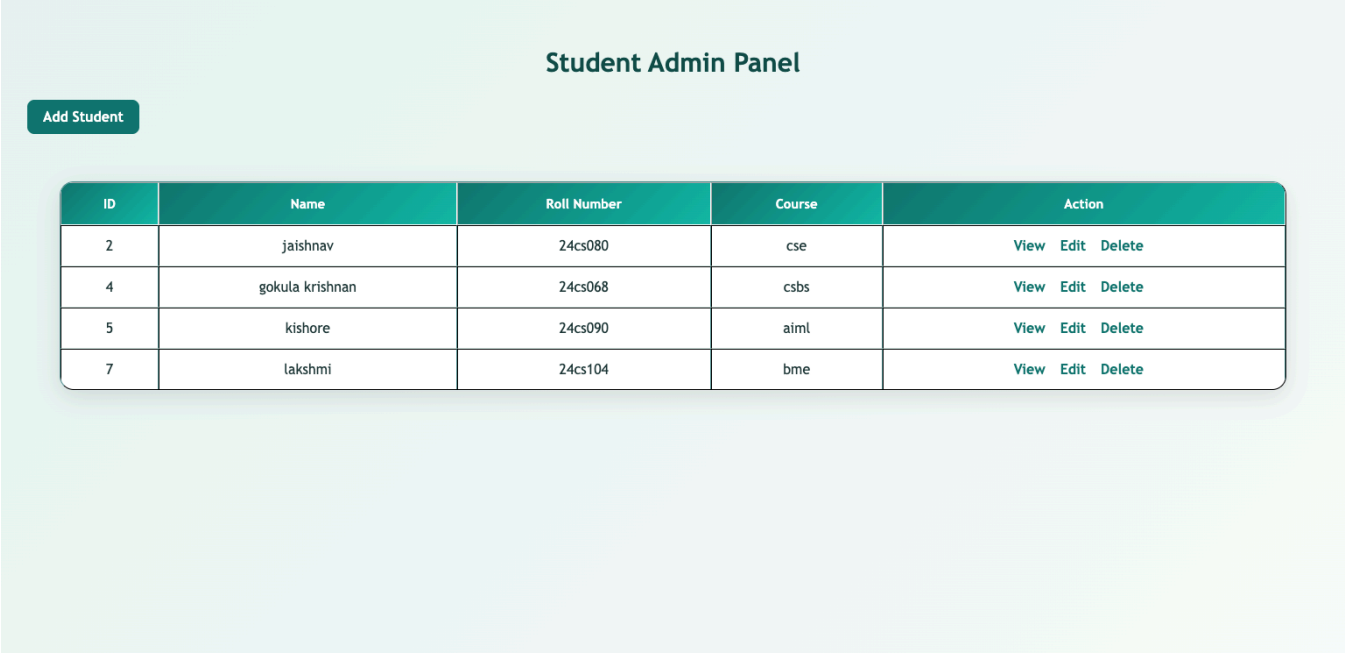
    table {
        width: 100%;
        font-size: 13px;
    }

    th,
    td {
        padding: 8px;
    }

    body > p {
        width: 90%;
    }

    img {
        max-width: 180px;
    }
}
```

Output


Student Admin Panel
Add Student
ID
Name
Roll Number
Course
Action
2
jaishnav
24cs080
cse
View Edit Delete
4
gokula krishnan
24cs068
csbs
View Edit Delete
5
kishore
24cs090
aiml
View Edit Delete
7
lakshmi
24cs104
bme
View Edit Delete

QUESTION 3

Question: Develop Servlet pages that connect to MySQL using JDBC, retrieve and stream images/documents, display student profiles dynamically in HTML/JSP with img tags, list multiple students and files, and ensure session handling for logged-in users.

Code

File: pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
           https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>portal</groupId>
    <artifactId>student-profile-portal</artifactId>
    <version>1.0.0</version>
    <packaging>war</packaging>
    <properties>
        <maven.compiler.release>17</maven.compiler.release>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>
    <dependencies>
        <dependency>
            <groupId>jakarta.servlet</groupId>
            <artifactId>jakarta.servlet-api</artifactId>
            <version>6.0.0</version>
            <scope>provided</scope>
        </dependency>
        <dependency>
            <groupId>com.mysql</groupId>
            <artifactId>mysql-connector-j</artifactId>
            <version>9.4.0</version>
        </dependency>
        <dependency>
            <groupId>org.owasp.encoder</groupId>
            <artifactId>encoder</artifactId>
            <version>1.3.1</version>
        </dependency>
        <dependency>
            <groupId>jakarta.servlet.jsp.jstl</groupId>
            <artifactId>jakarta.servlet.jsp.jstl-api</artifactId>
            <version>3.0.2</version>
        </dependency>
        <dependency>
            <groupId>org.glassfish.web</groupId>
            <artifactId>jakarta.servlet.jsp.jstl</artifactId>
            <version>3.0.1</version>
        </dependency>
    </dependencies>
    <build>
        <finalName>student-profile-portal</finalName>
    </build>
</project>
```

File: schema.sql

```
-- This project reads the student_admin schema created for Question 2.
-- Tables required: students(id, name, roll_no, email, course)
-- student_files(id, student_id, original_name, stored_name, mime_type, file_size, category,
  uploaded_at)
-- No duplicate schema is created here; use the schema.sql from Q2.
```

File: src/main/java/portal/AuthFilter.java

```
package portal;

import java.io.IOException;
import jakarta.servlet.Filter;
import jakarta.servlet.FilterChain;
import jakarta.servlet.ServletException;
import jakarta.servlet.ServletRequest;
import jakarta.servlet.ServletResponse;
import jakarta.servlet.annotation.WebFilter;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebFilter(urlPatterns = {"/students", "/media"})
public class AuthFilter implements Filter {
    @Override
    public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain)
            throws IOException, ServletException {
        HttpServletRequest req = (HttpServletRequest) request;
        HttpServletResponse res = (HttpServletResponse) response;
        HttpSession session = req.getSession(false);
        if (session == null || session.getAttribute("user") == null) {
            res.sendRedirect(req.getContextPath() + "/login.jsp");
            return;
        }
        chain.doFilter(request, response);
    }
}
```

File: src/main/java/portal/DBUtil.java

```
package portal;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public final class DBUtil {
    private static final String URL =
        "jdbc:mysql://localhost:3306/student_admin?useSSL=false&serverTimezone=UTC";
    private static final String USER = "root";
    private static final String PASSWORD = "";

    private DBUtil() { }

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

File: src/main/java/portal/LoginServlet.java

```
package portal;

import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {
```

```
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String username = request.getParameter("username");
        String password = request.getParameter("password");
        if (username == null || password == null || username.isBlank() || password.isBlank()) {
            request.setAttribute("error", "Enter both username and password.");
            request.getRequestDispatcher("/login.jsp").forward(request, response);
            return;
        }

        // Demo-only credentials. Replace with a users table and password hashing in
          production.
        if ("student".equals(username) && "Student@123".equals(password)) {
            HttpSession session = request.getSession(true);
            session.setAttribute("user", username);
            session.setMaxInactiveInterval(20 * 60);
            response.sendRedirect(request.getContextPath() + "/students");
            return;
        }
        request.setAttribute("error", "Invalid login details.");
        request.getRequestDispatcher("/login.jsp").forward(request, response);
    }
}
```

File: src/main/java/portal/LogoutServlet.java

```
package portal;

import java.io.IOException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
      IOException {
        HttpSession session = request.getSession(false);
        if (session != null) session.invalidate();
        response.sendRedirect(request.getContextPath() + "/login.jsp");
    }
}
```

File: src/main/java/portal/MediaServlet.java

```
package portal;

import java.io.IOException;
import java.io.InputStream;
import java.nio.file.Files;
import java.nio.file.Path;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/media")
public class MediaServlet extends HttpServlet {
    // Set this to the same private uploads directory used by the PHP application.
    private Path uploadRoot;
```

```
    @Override
    public void init() throws ServletException {
        String configured = getServletContext().getInitParameter("uploadRoot");
        if (configured == null || configured.isBlank()) {
            throw new ServletException("Configure the private uploadRoot context parameter.");
        }
        uploadRoot = Path.of(configured).toAbsolutePath().normalize();
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String fileId = request.getParameter("id");
        if (fileId == null || !fileId.matches("\\d+")) {
            response.sendError(HttpServletResponse.SC_BAD_REQUEST, "A valid file ID is
              required.");
            return;
        }
        String sql = "SELECT stored_name, original_name, mime_type FROM student_files WHERE id
          = ?";
        try (Connection connection = DBUtil.getConnection();
             PreparedStatement statement = connection.prepareStatement(sql)) {
            statement.setInt(1, Integer.parseInt(fileId));
            try (ResultSet result = statement.executeQuery()) {
                if (!result.next()) {
                    response.sendError(HttpServletResponse.SC_NOT_FOUND, "File not found.");
                    return;
                }
                String storedName = result.getString("stored_name");
                String mime = result.getString("mime_type");
                String originalName = result.getString("original_name");
                Path file = uploadRoot.resolve(storedName).normalize();
                if (!file.startsWith(uploadRoot) || !Files.isRegularFile(file)) {
                    response.sendError(HttpServletResponse.SC_NOT_FOUND, "Stored file not
                      found.");
                    return;
                }
                response.setContentType(mime);
                response.setHeader("X-Content-Type-Options", "nosniff");
                response.setHeader("Content-Disposition", "inline; filename=\"" +
                    originalName.replace("\"", "") + "\"");
                response.setContentLengthLong(Files.size(file));
                try (InputStream input = Files.newInputStream(file)) {
                    input.transferTo(response.getOutputStream());
                }
            }
        } catch (SQLException exception) {
            getServletContext().log("Could not retrieve uploaded file", exception);
            response.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR, "Unable to
              retrieve file.");
        }
    }
}
```

File: src/main/java/portal/StudentServlet.java

```
package portal;

import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
```

```
@WebServlet("/students")
public class StudentServlet extends HttpServlet {
    public static class Student {
        private final int id;
        private final String name, rollNo, email, course;
        public Student(int id, String name, String rollNo, String email, String course) {
            this.id=id; this.name=name; this.rollNo=rollNo; this.email=email;
              this.course=course;
        }
        public int getId(){return id;}
        public String getName(){return name;}
        public String getRollNo(){return rollNo;}
        public String getEmail(){return email;}
        public String getCourse(){return course;}
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        List<Student> students = new ArrayList<>();
        String sql = "SELECT id, name, roll_no, email, course FROM students ORDER BY name";
        try (Connection connection = DBUtil.getConnection();
             PreparedStatement statement = connection.prepareStatement(sql);
             ResultSet result = statement.executeQuery()) {
            while (result.next()) {
                students.add(new Student(result.getInt("id"), result.getString("name"),
                    result.getString("roll_no"), result.getString("email"),
                      result.getString("course")));
            }
        } catch (SQLException exception) {
            getServletContext().log("Could not retrieve student records", exception);
            response.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR, "Unable to load
              student records.");
            return;
        }
        request.setAttribute("students", students);
        request.getRequestDispatcher("/students.jsp").forward(request, response);
    }
}
```

File: src/main/webapp/WEB-INF/web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee" version="6.0">
    <display-name>Student Records Portal</display-name>
    <context-param>
        <param-name>uploadRoot</param-name>
        <param-value>/replace/with/your/private/uploads/folder</param-value>
    </context-param>
    <session-config>
        <session-timeout>20</session-timeout>
        <cookie-config>
            <http-only>true</http-only>
        </cookie-config>
        <tracking-mode>COOKIE</tracking-mode>
    </session-config>
    <welcome-file-list>
        <welcome-file>login.jsp</welcome-file>
    </welcome-file-list>
</web-app>
```

File: src/main/webapp/login.jsp

```
<%@ page contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
```

```
    <title>Student Portal Login</title>
    <link rel="stylesheet" href="style.css">
</head>
<body class="login-page">
    <main class="login-card">
        <div class="brand-mark">SA</div>
        <p class="eyebrow">STUDENT PORTAL</p>
        <h1>Welcome back</h1>
        <p class="muted">Sign in to view student profiles and documents.</p>

        <% if (request.getAttribute("error") != null) { %>
            <div class="alert">
                <%= org.owasp.encoder.Encode.forHtml(
                        String.valueOf(request.getAttribute("error"))) %>
            </div>
        <% } %>

        <form method="post" action="login" class="student-form">
            <label>Username
                <input name="username" required autocomplete="username">
            </label>
            <label>Password
                <input type="password" name="password"
                       required autocomplete="current-password">
            </label>
            <button class="button" type="submit">Sign in</button>
        </form>

        <p class="hint">Demonstration account: student / Student@123</p>
    </main>
</body>
</html>
```

File: src/main/webapp/students.jsp

```
<%@ page contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<%@ page import="java.util.List,portal.StudentServlet.Student" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Student Directory</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header class="topbar">
        <div>
            <span class="brand-mark small">SA</span>
            <strong>Student Records Portal</strong>
        </div>
        <form method="post" action="logout">Signed in · ${sessionScope.user}
            <button class="button secondary" type="submit">Sign out</button>
        </form>
    </header>

    <main class="container">
        <p class="eyebrow">DIRECTORY</p>
        <h1>Student Directory</h1>
        <p class="muted">Profile details and uploaded files fetched from MySQL.</p>

        <% List<Student> students = (List<Student>) request.getAttribute("students"); %>
        <% if (students == null || students.isEmpty()) { %>
            <section class="panel">
                <p>No student records are available.</p>
            </section>
        <% } else { %>
            <div class="student-grid">
                <% for (Student student : students) { %>
```

```
                    <article class="student-card">
                        <div class="avatar">
                            <%= org.owasp.encoder.Encode.forHtml(
                                    student.getName().substring(0,1).toUpperCase()) %>
                        </div>
                        <h2><%= org.owasp.encoder.Encode.forHtml(student.getName()) %></h2>
                        <p>
                            <span>Roll number</span>
                            <strong><%= org.owasp.encoder.Encode.forHtml(student.getRollNo())
                              %></strong>
                        </p>
                        <p>
                            <span>Email</span>
                            <strong><%= org.owasp.encoder.Encode.forHtml(student.getEmail())
                              %></strong>
                        </p>
                        <p>
                            <span>Course</span>
                            <strong><%= org.owasp.encoder.Encode.forHtml(student.getCourse())
                              %></strong>
                        </p>

                        <%-- Associated files are loaded by the file servlet
                             using file IDs from student_files. --%>
                        <%
                            try (java.sql.Connection connection =
                              portal.DBUtil.getConnection();
                                 java.sql.PreparedStatement statement =
                                   connection.prepareStatement(
                                     "SELECT id, original_name, mime_type, category "
                                      + "FROM student_files WHERE student_id=? "
                                      + "ORDER BY uploaded_at DESC")) {
                                statement.setInt(1, student.getId());
                                try (java.sql.ResultSet files = statement.executeQuery()) {
                                    boolean hasFiles = false;
                                    while (files.next()) {
                                        hasFiles = true;
                                        String mime = files.getString("mime_type");
                                        String fileName = org.owasp.encoder.Encode.forHtml(
                                                files.getString("original_name"));
                                        int fileId = files.getInt("id");
                                        if (mime != null && mime.startsWith("image/")) {
                        %>
                                            <img class="profile-image" src="media?id=<%= fileId
                                              %>"
                                                 alt="Uploaded student image">
                        <%
                                        }
                        %>
                                        <a class="file-link" href="media?id=<%= fileId %>"
                                           target="_blank" rel="noopener"><%= fileName %></a>
                        <%
                                    }
                                    if (!hasFiles) {
                        %>
                                        <p class="muted">No uploaded files.</p>
                        <%
                                    }
                                }
                            } catch (java.sql.SQLException exception) {
                                application.log("Could not list student files", exception);
                        %>
                                <p class="muted">Files are temporarily unavailable.</p>
                        <%
                            }
                        %>
                    </article>
                <% } %>
            </div>
        <% } %>
```

```
    </main>

    <footer>Web Technology · Gokula Krishnan R · 24CS068 · III-CSE-B</footer>
</body>
</html>
```

File: src/main/webapp/style.css

```
:root {
    --navy: #134e4a;
    --blue: #0d9488;
    --bg: #f4f8f7;
    --ink: #1f3a3a;
    --muted: #6b7f7d;
    --border: #d9e8e5;
}

* {
    box-sizing: border-box;
}

body {
    margin: 0;
    background: var(--bg);
    color: var(--ink);
    font-family: "Segoe UI", Arial, sans-serif;
    font-size: 12pt;
    line-height: 1.5;
}

.topbar {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 16px max(5vw, 20px);
    background: var(--navy);
    color: white;
}

.topbar > div {
    display: flex;
    align-items: center;
    gap: 12px;
}

.brand-mark {
    display: grid;
    place-items: center;
    width: 58px;
    height: 58px;
    border-radius: 16px;
    background: #0d94881a;
    color: var(--blue);
    font-size: 20pt;
    font-weight: bold;
}

.brand-mark.small {
    width: 40px;
    height: 40px;
    background: #ffffff20;
    color: white;
    font-size: 14pt;
}

.container {
    width: min(1150px, 92%);
    margin: 34px auto 50px;
}
```

```
h1 {
    margin: 0 0 8px;
    color: var(--navy);
    font-size: 25pt;
}

.eyebrow {
    margin: 0 0 5px;
    color: var(--blue);
    font-size: 9pt;
    font-weight: bold;
    letter-spacing: 1.3px;
}

.muted,
.hint {
    color: var(--muted);
    font-size: 10pt;
}

.panel,
.student-card,
.login-card {
    background: white;
    border: 1px solid var(--border);
    border-radius: 18px;
    padding: 22px;
    box-shadow: 0 8px 28px #0f3d3a12;
}

.student-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(270px, 1fr));
    gap: 18px;
    margin-top: 22px;
}

.student-card h2 {
    margin: 10px 0;
    color: var(--navy);
    font-size: 17pt;
}

.avatar {
    display: grid;
    place-items: center;
    width: 48px;
    height: 48px;
    border-radius: 14px;
    background: #d5efe9;
    color: var(--navy);
    font-size: 19pt;
    font-weight: bold;
}

.student-card p {
    display: grid;
    gap: 2px;
    padding: 8px 0;
    margin: 0;
    border-bottom: 1px solid #edf1f7;
    overflow-wrap: anywhere;
}

.student-card p span {
    color: var(--muted);
    font-size: 10pt;
}

.profile-image {
```

```
    display: block;
    max-width: 100%;
    max-height: 220px;
    object-fit: contain;
    margin: 12px 0;
    border-radius: 8px;
}

.file-link {
    display: block;
    margin: 8px 0;
    overflow-wrap: anywhere;
}

.button {
    border: 0;
    border-radius: 8px;
    padding: 10px 16px;
    background: var(--blue);
    color: white;
    font: inherit;
    font-weight: bold;
    cursor: pointer;
}

.button.secondary {
    background: #ffffff20;
    color: white;
    border: 1px solid #ffffff66;
}

.login-page {
    min-height: 100vh;
    display: grid;
    place-items: center;
    padding: 20px;
    background: linear-gradient(140deg, #e6f4f1, #f7fbfa);
}

.login-card {
    width: min(440px, 100%);
    padding: 32px;
}

.login-card h1 {
    margin-top: 8px;
}

.login-card .brand-mark {
    margin-bottom: 18px;
}

.student-form {
    display: grid;
    gap: 15px;
    margin-top: 22px;
}

.student-form label {
    display: grid;
    gap: 6px;
    font-weight: bold;
}

.student-form input {
    padding: 11px;
    border: 1px solid #c5dcd8;
    border-radius: 8px;
    font: inherit;
}
```

```
.student-form input:focus {
    outline: 3px solid #0d948826;
    border-color: var(--blue);
}

.alert {
    padding: 10px;
    border-radius: 8px;
    background: #fff0f0;
    color: #9e2424;
    margin-top: 12px;
}

footer {
    text-align: center;
    color: var(--muted);
    padding: 20px;
}

@media (max-width: 600px) {
    .login-card {
        padding: 22px;
    }
    .topbar {
        padding: 14px 4%;
    }
    .container {
        margin-top: 24px;
    }
}
```

Output

